

**AN INDUSTRY ORIENTED MINI PROJECT REPORT
ON**

Voice Controlled Home Automation Using Bluetooth

Submitted in partial fulfillment of the requirements for the award of

**BACHELOR OF TECHNOLOGY
IN
ELECTRONICS & COMMUNICATION ENGINEERING**



By

V. YASHVANTH (23281A0480)

R. NIKHIL (24285A0429)

N.BHAVANI (24285A0427)

D.SAI MEGHANA (24285A0435)

B.SAI KIRAN (23281A0483)

Under the guidance of

Mrs.N.SUMA

Assistant Professor

**KAMALA INSTITUTE OF TECHNOLOGY & SCIENCE
SINGAPUR, HUZURABAD, KARIMNAGAR, TELANGANA,INDIA – 505 468**

Approved by AICTE-New Delhi and Affiliated to JNTU-HYDERABAD,TS

Accredited by NAAC With A++,Accredited by NBA

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2025-2026

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

CERTIFICATE

This is to certify that this is a bonafide record of the project work entitled “**Voice Controlled Home Automation Using Bluetooth**” is a bonafide work carried out by **V.YASHVANTH (23281A0480), R.NIKHIL (23281A0429), N.BHAVANI (24285A0427), D.SAI MEGHANA (24285A0435), B.SAIKIRAN (23281A0483)** in partial fulfillment of the requirements for the degree **BACHELOR OF TECHNOLOGY** in **ELECTRONICS & COMMUNICATION ENGINEERING** by the Jawaharlal Nehru Technological University,Hyderabad,during the academic year 2025-2026

Project Guide

Mrs.N.SUMA

Assistant Professor

Project Coordinator

DR.P.SURESH

Assistant Professor

Head of the Department

Dr.B.RAMESH

Professor



VEGA PLACEMENTS & TRAINING

A Better Gate way to Getaways ...

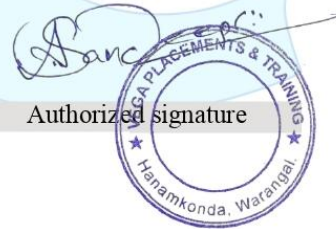
PROJECT COMPLETION CERTIFICATE

This is to certify that V. YASHVANTH(23281A0480),N.BHAVANI(24285A0427),

D.SAI MEGHANA (24285A0435),B.SAI KIRAN(23281A0483),R. NIKHIL(24285A0429) are the students of **Kamala Institute of Technology and Science, Huzurabad**, Studying in the course of B.Tech (Electronics and Communication Engineering) completed Project work on “VOICE CONTROLLED HOME AUTOMATION USING BLUETOOTH”, **B.tech** Degree during the period **09-02-2026** to **02-03-2026** under our supervision. a part of our research unit. We found them extremely inquisitive and hardworking. Association with us was very truthful and we wish all the Best for their future endeavors.

Date: 02-03-2026

Authorized signature



Address Details: Vega Placements & Training, Sri Sai Pooja Complex, Kishanpura, Hanamkonda, Warangal, 506001.

Mobile no: 9948457532, 7416219336.

Email Id: vegaatwgl@gmail.com.

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V. YASHVANTH	23281A0480
R. NIKHIL	24285A0429
N.BHAVANI	24285A0427
D.SAI MEGHANA	24285A0434
B. SAI KIRAN	23281A0483

ABSTRACT

This project focuses on designing a Bluetooth-based voice controlled home automation system that allows users to control household appliances using simple voice commands. Instead of using a separate voice recognition module, the system makes use of a smartphone's built-in voice recognition, which makes the project cost-effective and easy to implement. When the user speaks a command in any language through the mobile application, the voice is converted into text and sent to the controller via Bluetooth communication. The microcontroller receives this command and activates the corresponding appliance through relay circuits. Devices such as lights and fans can be turned ON or OFF without any physical interaction.

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CHAPTER-1

INTRODUCTION

In recent years, home automation has gained significant importance due to rapid advancements in embedded systems, wireless communication, and smart devices. Home automation systems aim to provide comfort, energy efficiency, and improved control of household appliances. With the increasing use of smartphones, wireless technologies such as Bluetooth have become popular for short-range communication in automation applications.

This project focuses on the design and implementation of a **Bluetooth-based home automation system with multilingual voice command control**. The system allows users to control electrical appliances such as lights and fans using voice commands through a mobile phone. Unlike conventional systems that rely on manual switches or single-language input, this project supports voice commands in **English, Telugu, and Hindi**, making it more user-friendly and accessible.

The voice commands given through a mobile application are converted into text and transmitted to the Arduino microcontroller using a Bluetooth module. The microcontroller processes these commands and controls the respective appliances accordingly. An LCD is used to display the current status of the devices, providing real-time feedback to the user. The proposed system is low cost, easy to implement, and suitable for elderly and physically challenged people who may find traditional switches difficult to use.

This project demonstrates an effective approach to smart home automation using Bluetooth communication and multilingual voice recognition.

CHAPTER-2

CONSTRUCTION AND WORKING

2.1 CONSTRUCTION

The proposed voice-based home automation system is constructed using an Arduino Uno as the main control unit, a Bluetooth module (HC-05) for wireless communication, a 16×2 LCD display for status indication, a relay module for switching appliances, and household loads such as a light bulb and a fan. A mobile phone is used as the input device to provide voice commands, which are converted into text by a mobile application and transmitted to the Arduino through Bluetooth.

The Arduino Uno acts as the brain of the system. It receives the command data from the Bluetooth module, processes it, and generates the required control signals for the relay module. The HC-05 Bluetooth module is connected to the serial pins of the Arduino and enables short-range wireless communication between the mobile phone and the controller. The LCD display is connected to the Arduino to show the current status of the appliances, such as “Light ON” or “Fan OFF,” thereby giving real-time feedback to the user.

The relay module is an important part of the construction because the Arduino cannot directly operate high-voltage AC appliances. The relay serves as an electrically isolated switch between the low-voltage Arduino circuit and the AC-powered devices. The fan and bulb are connected through the relay outputs so that they can be safely turned ON or OFF according to the received voice command.

The entire control circuit is powered by a 5V DC supply, while the fan and light use the normal AC mains supply. The complete hardware arrangement is built in such a way that the user can control appliances without touching physical switches. This makes the system especially useful for elderly people, physically challenged users, and those who prefer a smart and convenient method of operating home devices.

The multilingual voice feature further improves the construction by allowing commands in English, Telugu, and Hindi.

2.2 WORKING

The working of the system begins when the user gives a voice command through a mobile phone. The mobile application recognizes the spoken words and converts them into a text-based command. This command is then sent wirelessly through the Bluetooth connection to the HC-05 module connected to the Arduino. After receiving the command, the Arduino reads and interprets it to determine which appliance should be controlled.

If the command corresponds to switching ON the light, the Arduino sends an output signal to the relay channel connected to the light. The relay then closes its switching contact and allows current to flow to the bulb, turning it ON. Similarly, if the user gives a command to switch OFF the light, the Arduino deactivates the corresponding relay, disconnecting the supply and turning the bulb OFF.

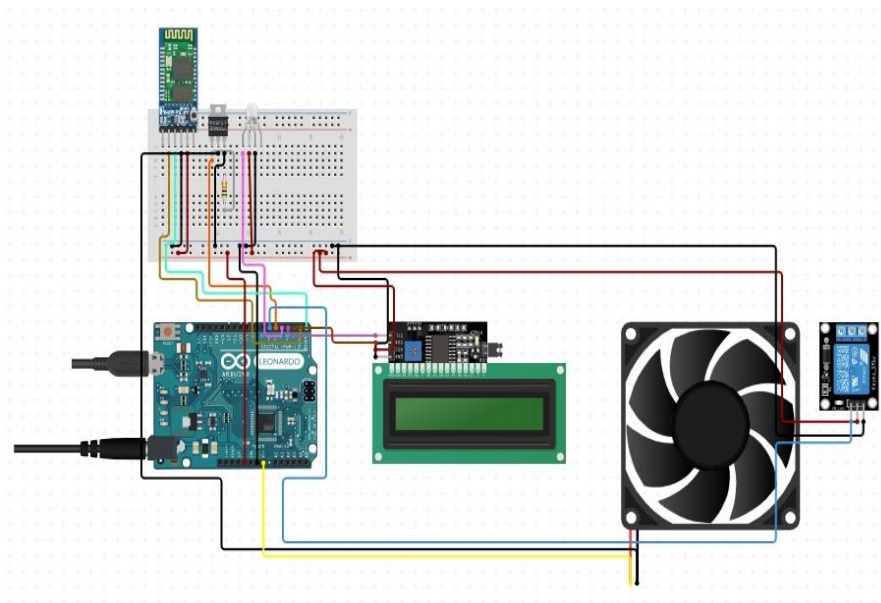


Fig:2.1 Voice controlled home automation using bluetooth circuit

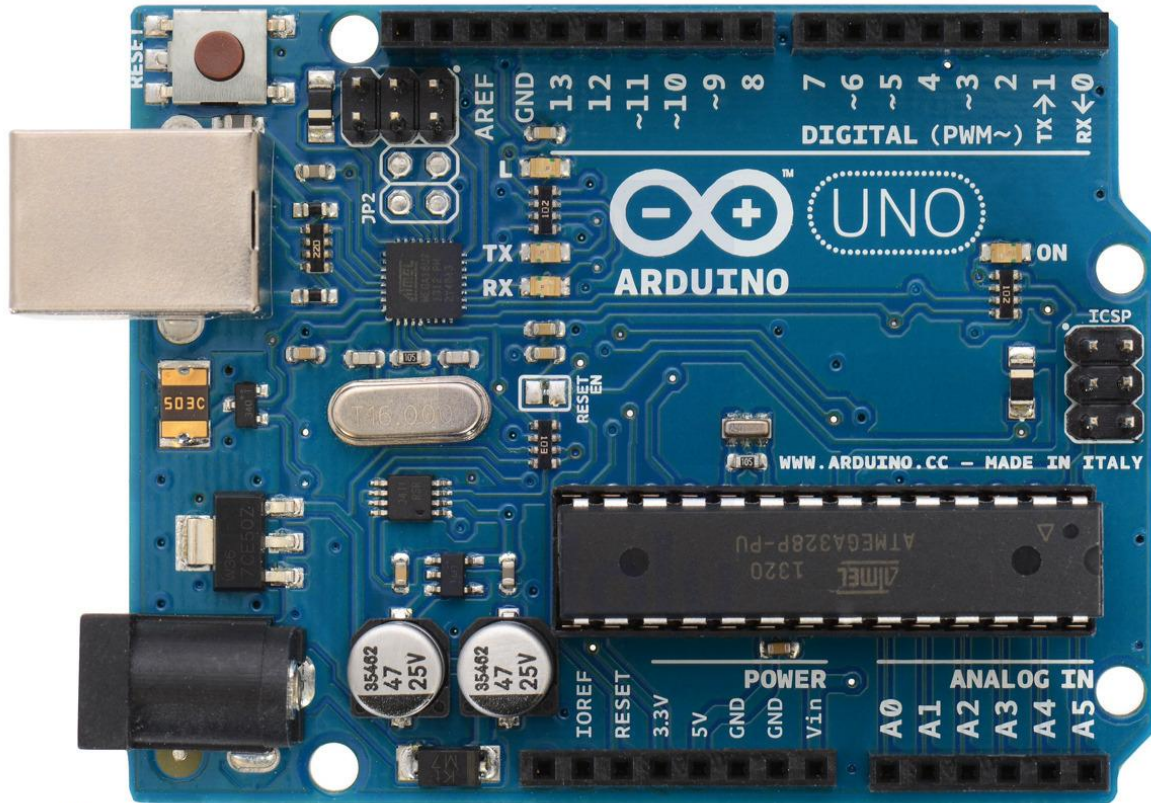
The same process is followed for controlling the fan. Depending on the spoken instruction, the Arduino activates or deactivates the relay connected to the fan. At the same time, the Arduino updates the LCD display with the current appliance status. For example, when the light is switched ON, the LCD shows “Light ON,” and when the fan is turned OFF, it displays “Fan OFF.”

This provides immediate confirmation to the user that the command has been successfully executed. The system works within the Bluetooth operating range, typically around 10 meters, making it suitable for indoor home automation applications.

CHAPTER-3

HARDWARE COMPONENTS

3.1 ARDUINO UNO:



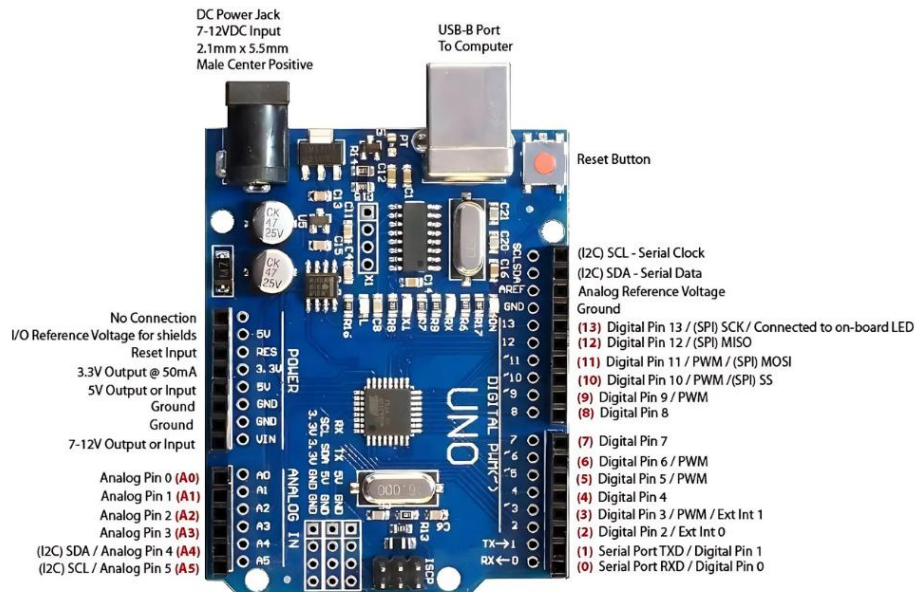
www.pololu.com

Fig 3.1:Arduino UNO board

The **Arduino Uno R3** is the main controller used in the voice-based home automation system. It acts as the brain of the entire project and controls all the connected devices.

The Arduino Uno is based on the **ATmega328P microcontroller**, which processes the commands received from the Bluetooth module. It receives input data, analyzes the command, and then activates the appropriate relay to control home appliances such as lights and fans.

The board contains **14 digital input/output pins**, **6 analog input pins**, a **USB interface**, and a **power jack**. In this project, the digital pins are used to control the relay module, while the serial communication pins are used for communication with the Bluetooth module.



Red numbers in paranthesis are the name to use when referencing that pin.
Analog pins are references as A0 thru A5 even when using as digital I/O

Fig 3.2:Arduino Uno Pin Diagram.

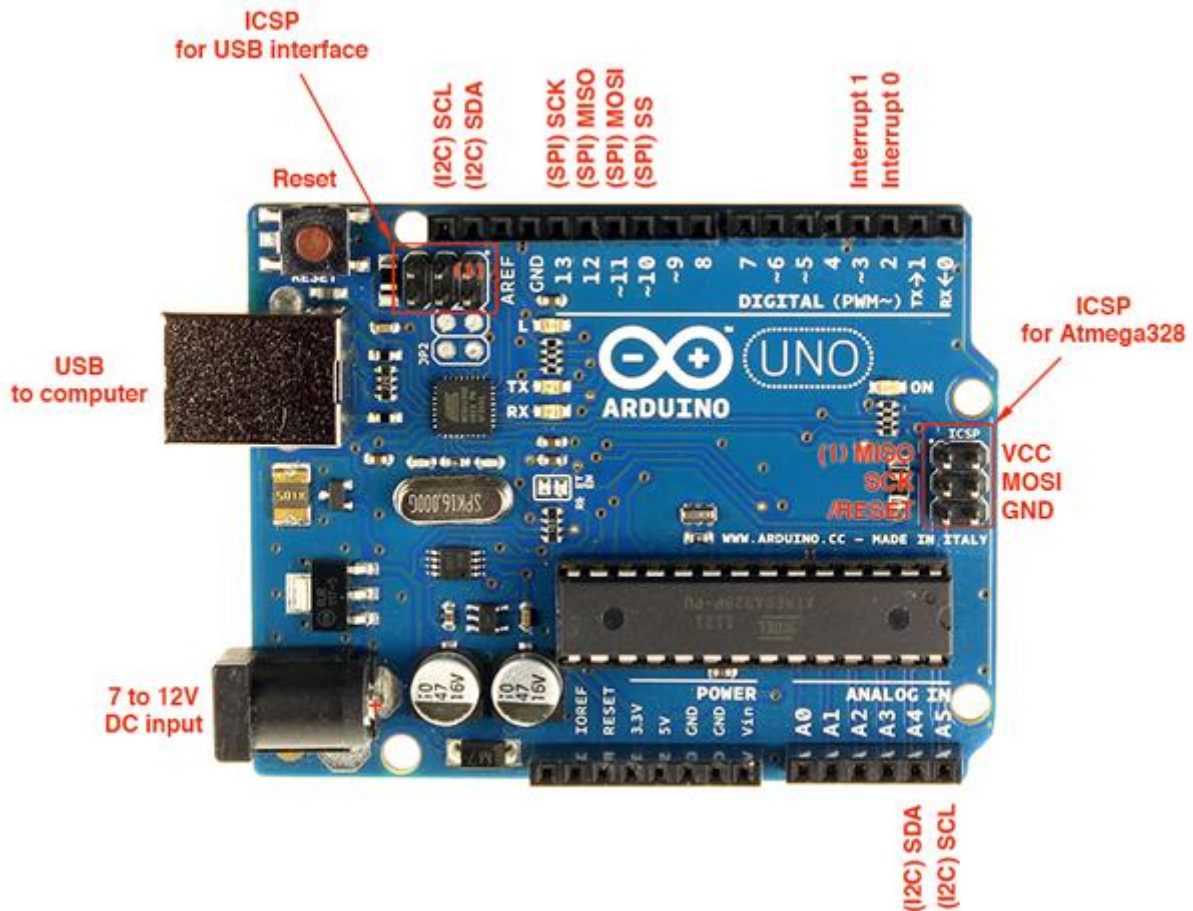


Fig 3.3:Arduino Uno board with pin labels.

The Arduino is programmed using the **Arduino IDE**, which allows the developer to write code in a simple C/C++ based language. Once programmed, the Arduino continuously listens for commands and executes the required operation.

3.2 Bluetooth Module (HC-05)



Fig 3.4:HC-05 Bluetooth module

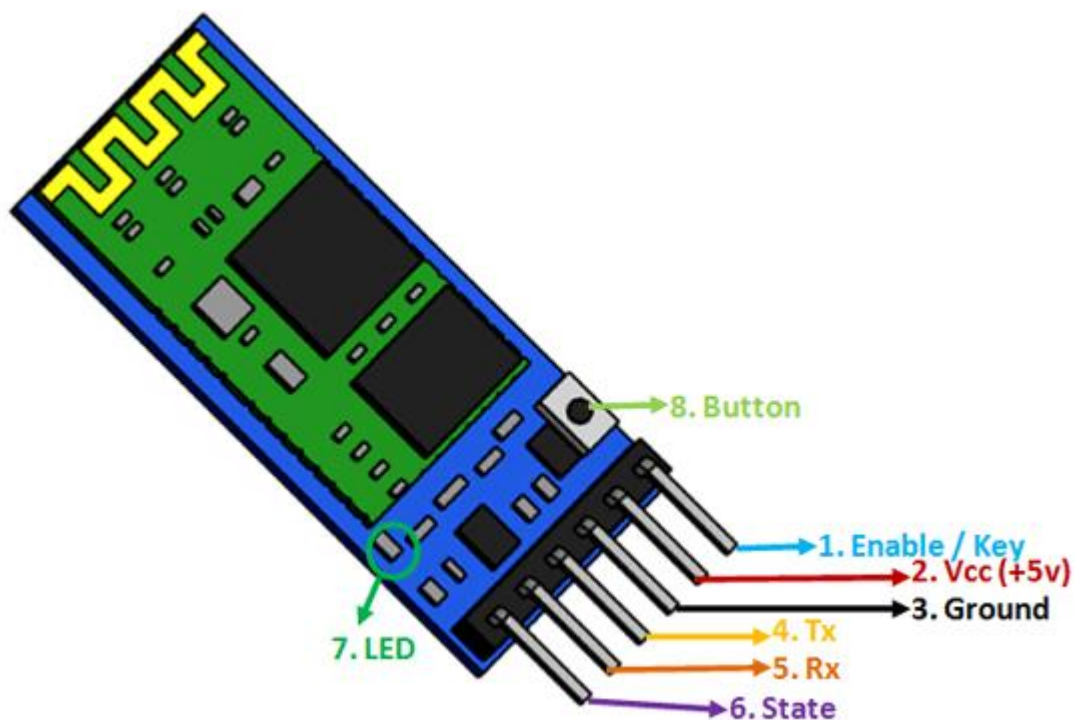


Fig 3.5:HC-05 Bluetooth module pin diagram.

The **HC-05 Bluetooth Module** is used for wireless communication between the mobile phone and the Arduino board.

The module allows the system to receive commands from the user's smartphone through Bluetooth technology. When the user speaks a command using the mobile application, the app converts the voice into text and sends it to the Bluetooth module.

The HC-05 module operates on the **Serial Communication Protocol (UART)** and communicates with the Arduino using the **TX (Transmit)** and **RX (Receive)** pins.

The typical operating range of the HC-05 module is about **10 meters**, which is sufficient for indoor home automation applications. It works with a **5V power supply** and is widely used because it is inexpensive, reliable, and easy to interface with microcontrollers

3.3 RELAY MODULE

The **Relay Module** is used to control high-voltage electrical appliances using the low-voltage output from the Arduino.

Since the Arduino operates at **5V**, it cannot directly control household devices that operate on **220V AC power**. The relay acts as an electrically isolated switch that connects or disconnects the AC supply to the appliance.

When the Arduino sends a HIGH signal to the relay module, the relay activates and closes the circuit, allowing current to flow to the connected appliance such as a light bulb or fan. When the signal is LOW, the relay turns OFF and the appliance stops working.

The relay module provides electrical isolation between the microcontroller and the high-voltage circuit, ensuring safety and protecting the Arduino from damage.



Fig 3.6:Single-channel 5V relay module.

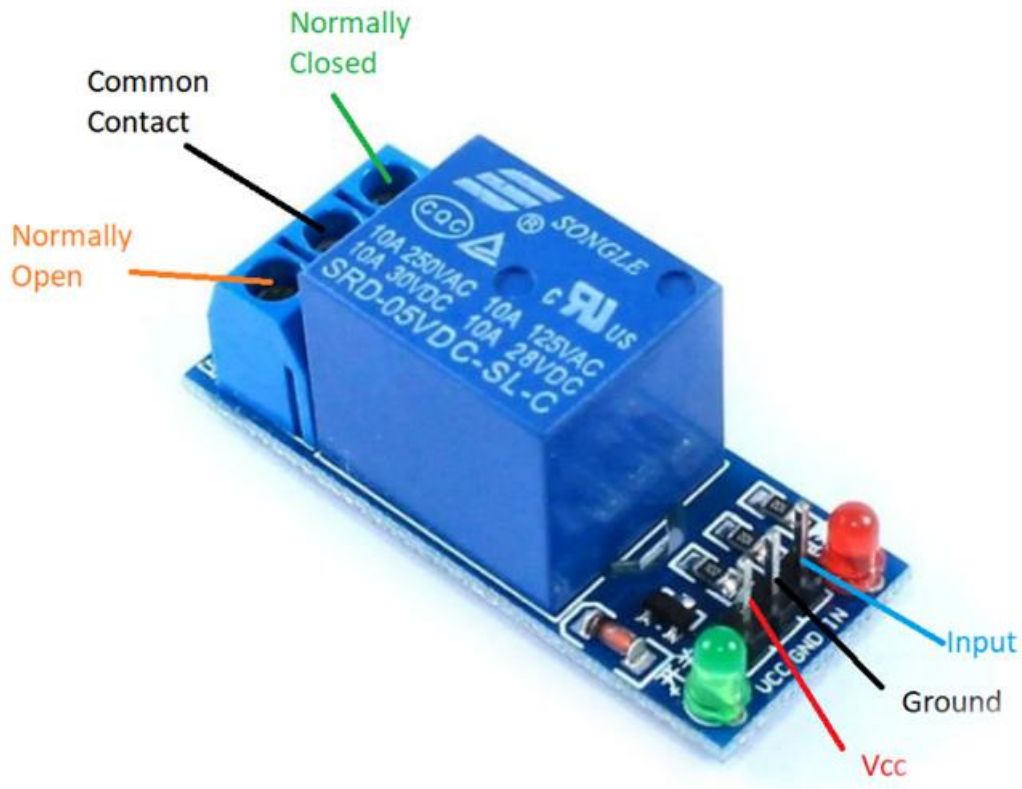


Fig 3.7:5V relay module pin and contact diagram

3.4 LCD DISPLAY (16×2)

The **16×2 LCD Display** is used to display system information and appliance status.

The display contains **two rows and sixteen columns**, allowing it to show simple messages such as “Light ON”, “Fan OFF”, or “System Ready”. It helps the user understand the current condition of the devices in the system.

The LCD can be connected directly using multiple data pins or through an **I2C module**, which reduces the number of wires required. In this project, the LCD provides visual feedback whenever a command is executed.

The display operates at **5V** and is widely used in embedded systems because it is simple, reliable, and easy to program.

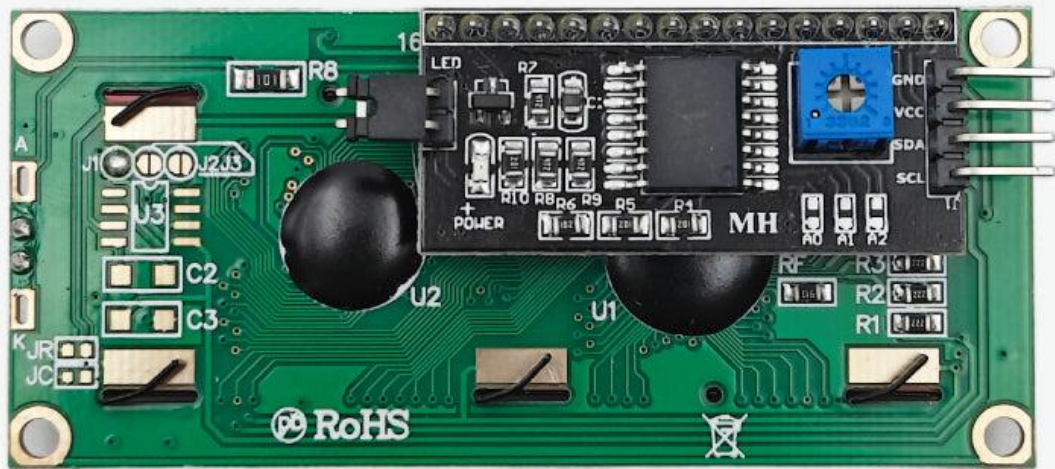


Fig 3.8:16×2 LCD display module with I2C backpack

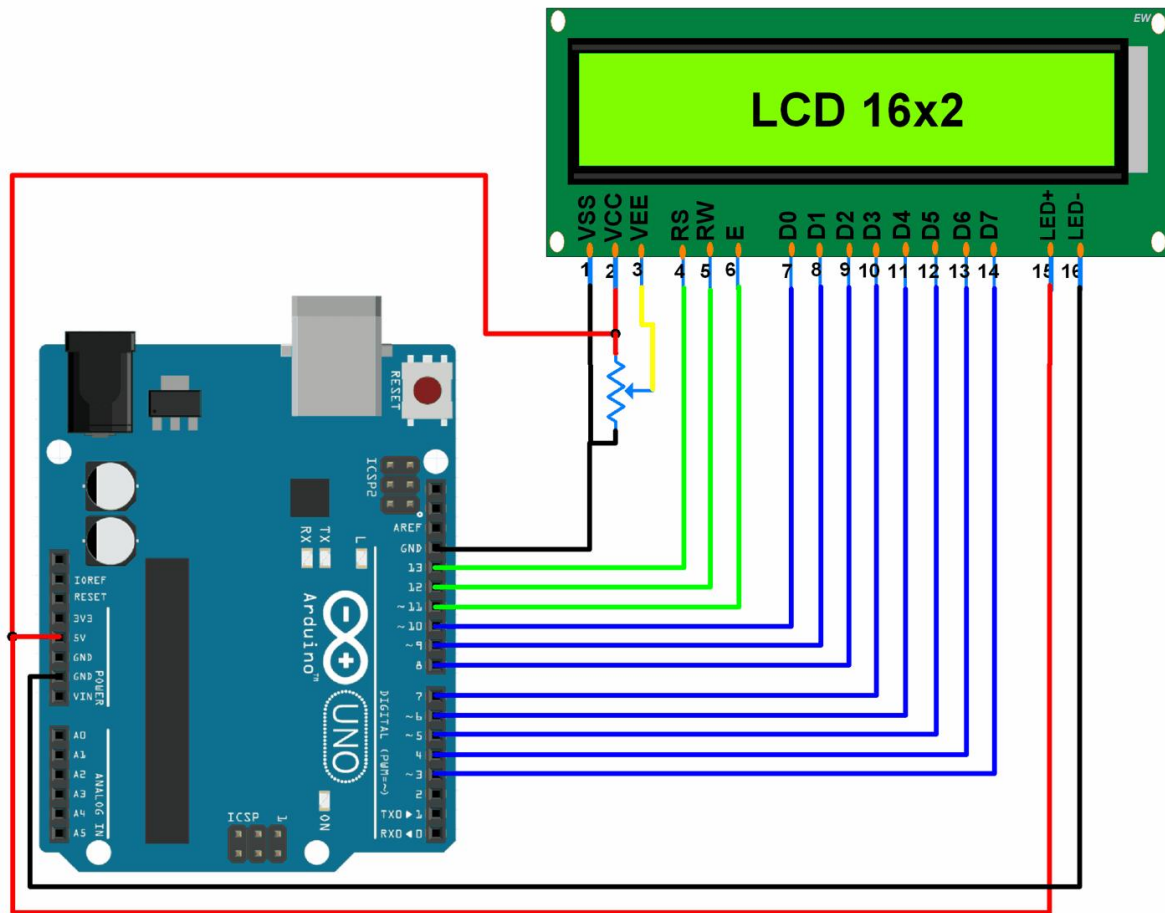


Fig 3.9:Interfacing of 16×2 LCD with Arduino Uno

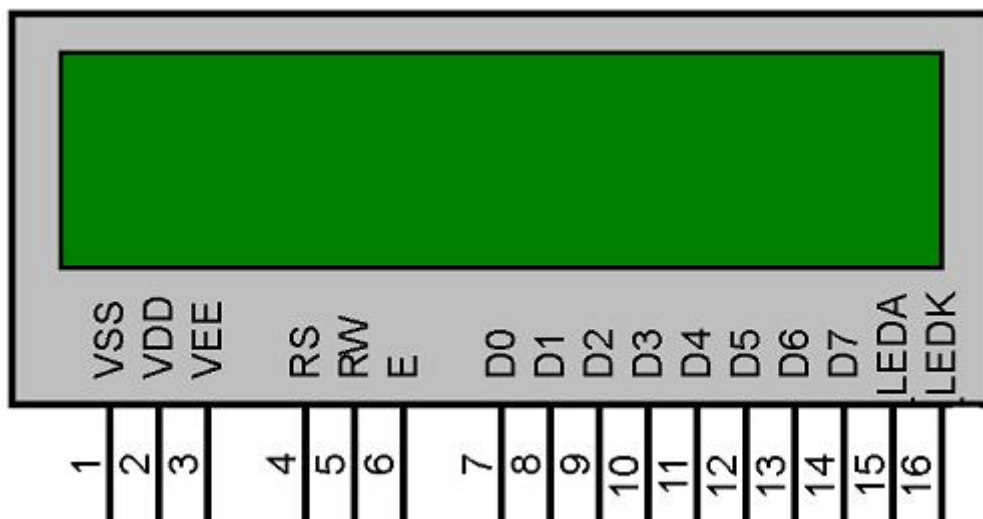


Fig 3.10:Pin Diagram of 16×2 LCD Display

3.5 LIGHT BULB

The **LED Light Bulb** is used as an example appliance in the home automation system.

The bulb represents a household electrical device that can be controlled using voice commands. When the user gives the command “**Light ON**”, the Arduino activates the relay module, allowing electricity to reach the bulb and turn it ON.

Similarly, when the command “**Light OFF**” is given, the relay disconnects the power supply and the bulb turns OFF.

This demonstrates how home lighting can be automated and controlled remotely using voice commands.

3.6 DC FAN



Fig 3.11: Brushless DC Cooling Fan

The **DC Fan** is another appliance used to demonstrate the functionality of the home automation system.

The fan is connected through the relay module so that it can be turned ON or OFF using voice commands. When the user gives the command “**Fan ON**”, the Arduino activates the relay channel connected to the fan, allowing current to flow and start the fan.

When the command “**Fan OFF**” is received, the Arduino deactivates the relay, cutting the power supply and stopping the fan.

Using the fan in the project helps demonstrate how multiple home appliances can be controlled using a single voice-based automation system.

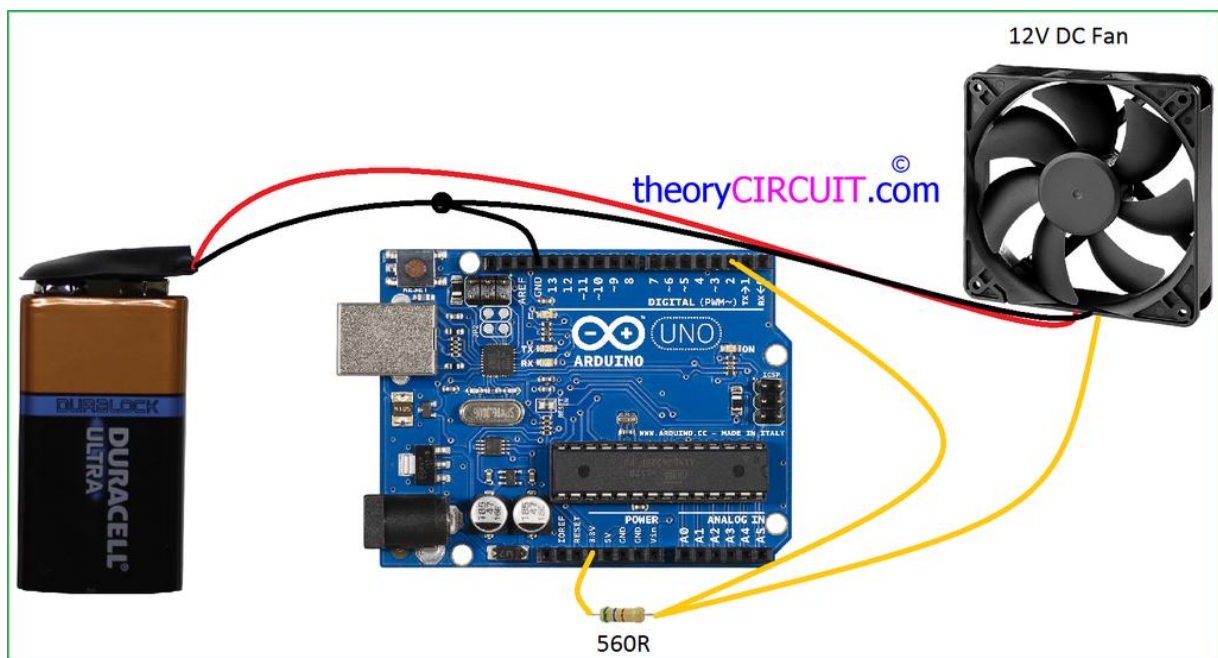


Fig 3.12: Interfacing of 12V DC Fan with Arduino Uno

CHAPTER-4

EMBEDDED SOFTWARE

Arduino is an open-source prototyping platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board.

To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing. Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments.

All Arduino boards are completely open- source, empowering users to build them independently and eventually adapt them to their particular needs. The software, too, is open-source, and it is growing through the contributions of users worldwide.

4.1 WHY ARDUINO?

Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics.

Designers and architects build interactive prototypes, musicians and artists use it for installations and to experiment with new musical instruments. Makers, of course, use it to build many of the projects exhibited at the Maker Faire, for example. Arduino is a key tool to learn new things. Anyone - children, hobbyists, artists, programmers - can start tinkering just

following the step by step instructions of a kit, or sharing ideas online with other members of the Arduino community.

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Net media's BX-24, Phidgets, MIT's Handyboard, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems:

- Inexpensive - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than \$50
- Cross-platform - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- Simple, clear programming environment - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- Open source and extensible software - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.
- Open source and extensible hardware - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version
- Of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money
- Getting Started with Arduino and Genuino products:-

Install the Arduino Software (IDE) on Windows PCs-

This document explains how to install the Arduino Software (IDE) on Windows machines.

- ✓ Download the Arduino Software(IDE)
- ✓ Proceed with board specific instructions

4.2 HOW TO DOWNLOAD THE ARDUINO SOFTWARE (IDE):

Get the latest version from the download page. You can choose between the Installer (.exe) and the Zip packages. We suggest you use the first one that installs directly everything you need to use the Arduino Software (IDE), including the drivers. With the Zip package you need to install the drivers manually.

When the download finishes, proceed with the installation and please allow the driver installation process when you get a warning from the operating system.

INSTALLATION:

In this section, we will learn in easy steps, how to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1 – First you must have your Arduino board (you can choose your favorite board) and a USB cable. In case you use Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.



Fig 4.1:USB cable

In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image.

Step 2 – Download Arduino IDE Software.

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your

operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

Step 3 – Power up your board.

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks.

Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4 – Launch Arduino IDE.

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

Step 5 – Open your first project.

Once the software starts, you have two options –

- Create a new project.
- Open an existing project example.

To create a new project, select File → **New**.

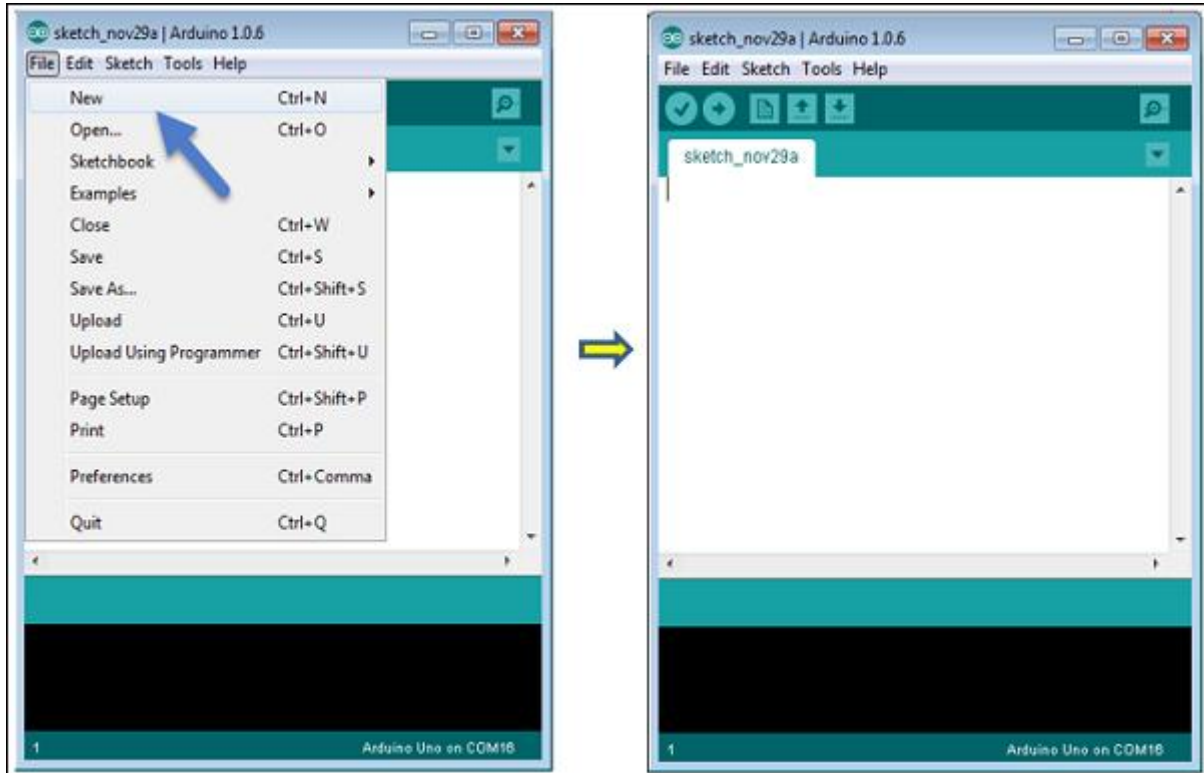


Fig 4.2: Creating a New Sketch in the Arduino IDE

To open an existing project example, select File → Example → Basics → Blink

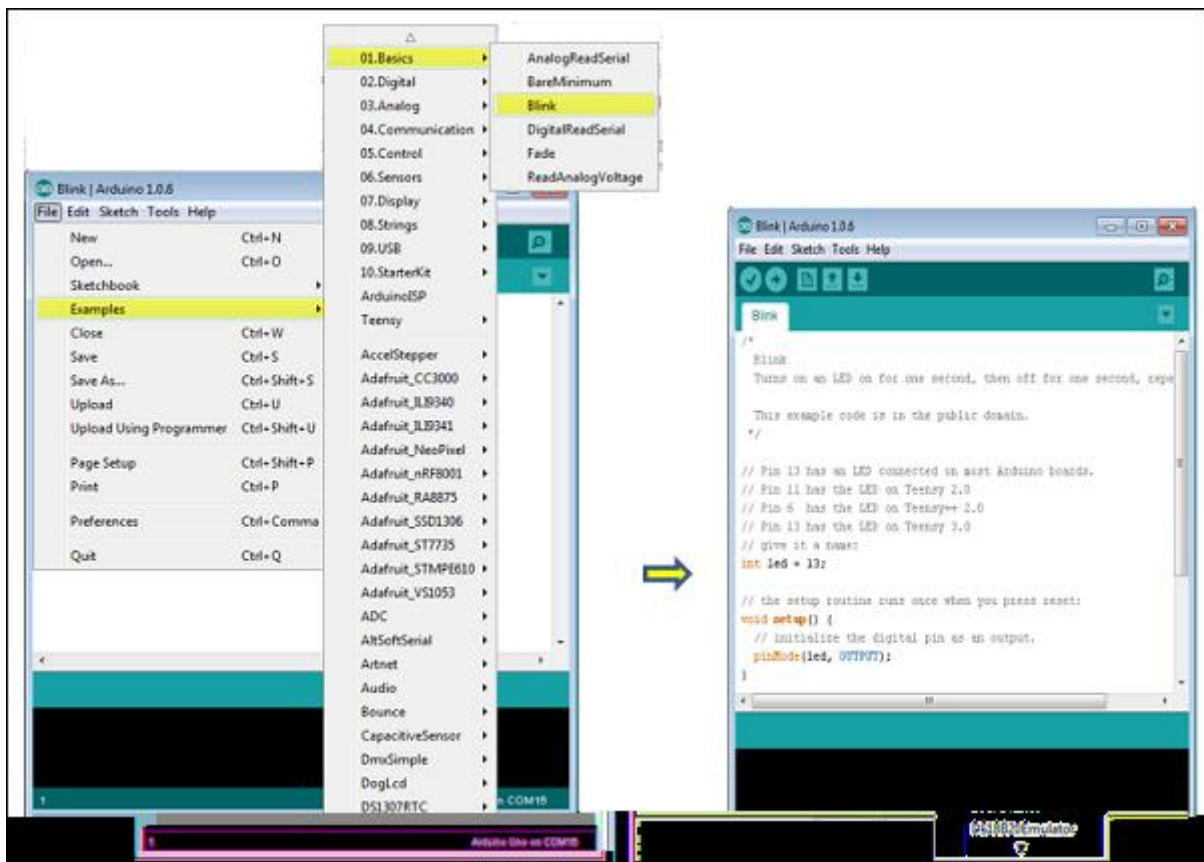


Fig 4.3: Selecting the Blink Example Program in the Arduino IDE

Here, we are selecting just one of the examples with the name **Blink**. It turns the LED on and off with some time delay. You can select any other example from the list.

Step 6 – Select your Arduino board.

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

Go to Tools → Board and select your board

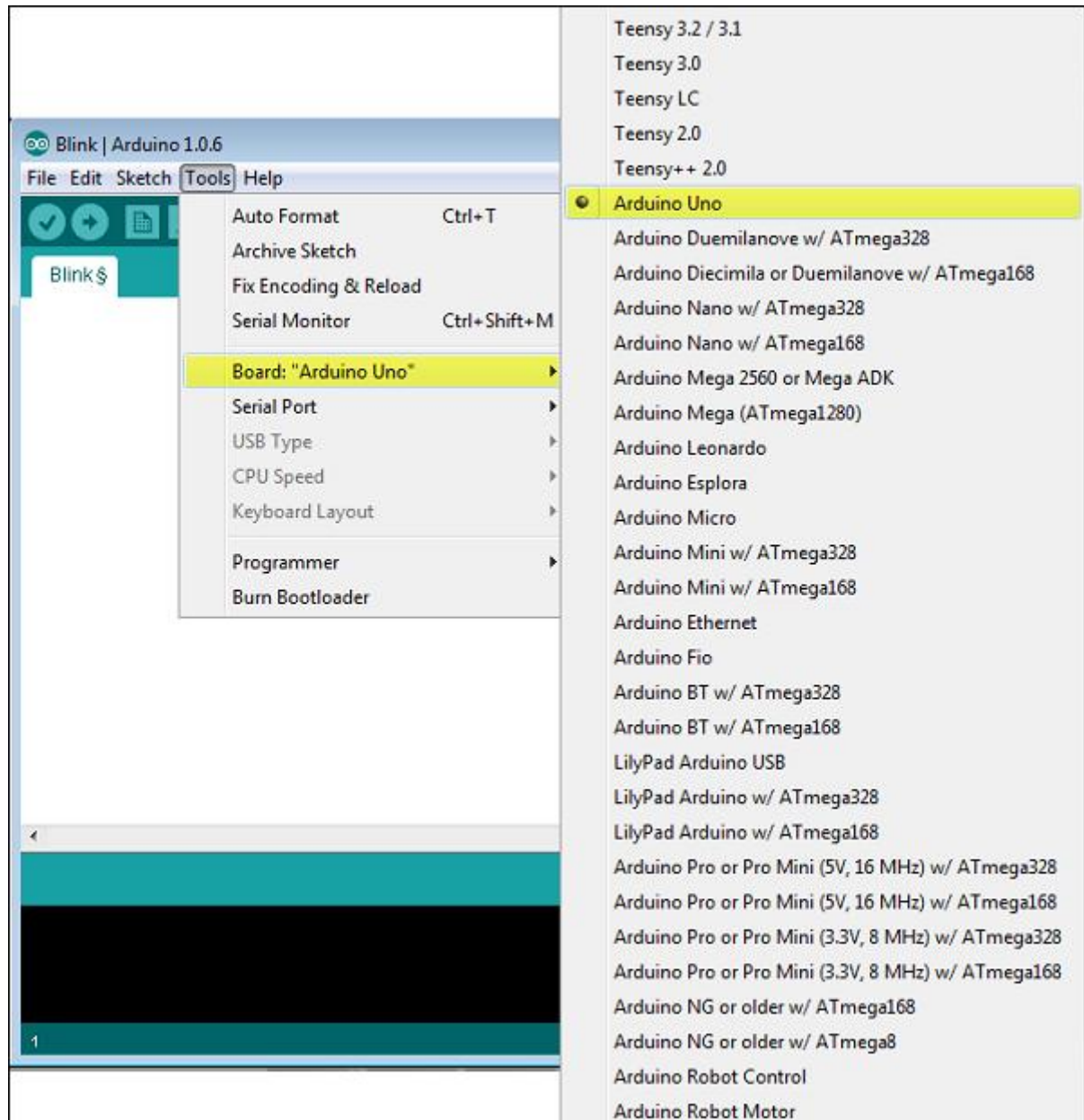


Fig 4.4: Selecting Arduino Uno Board in the Arduino IDE

Here, we have selected Arduino Uno board according to our tutorial, but you must select the name matching the board that you are using.

Step 7 – Select your serial port.

Select the serial device of the Arduino board. Go to **Tools** → **Serial Port** menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.

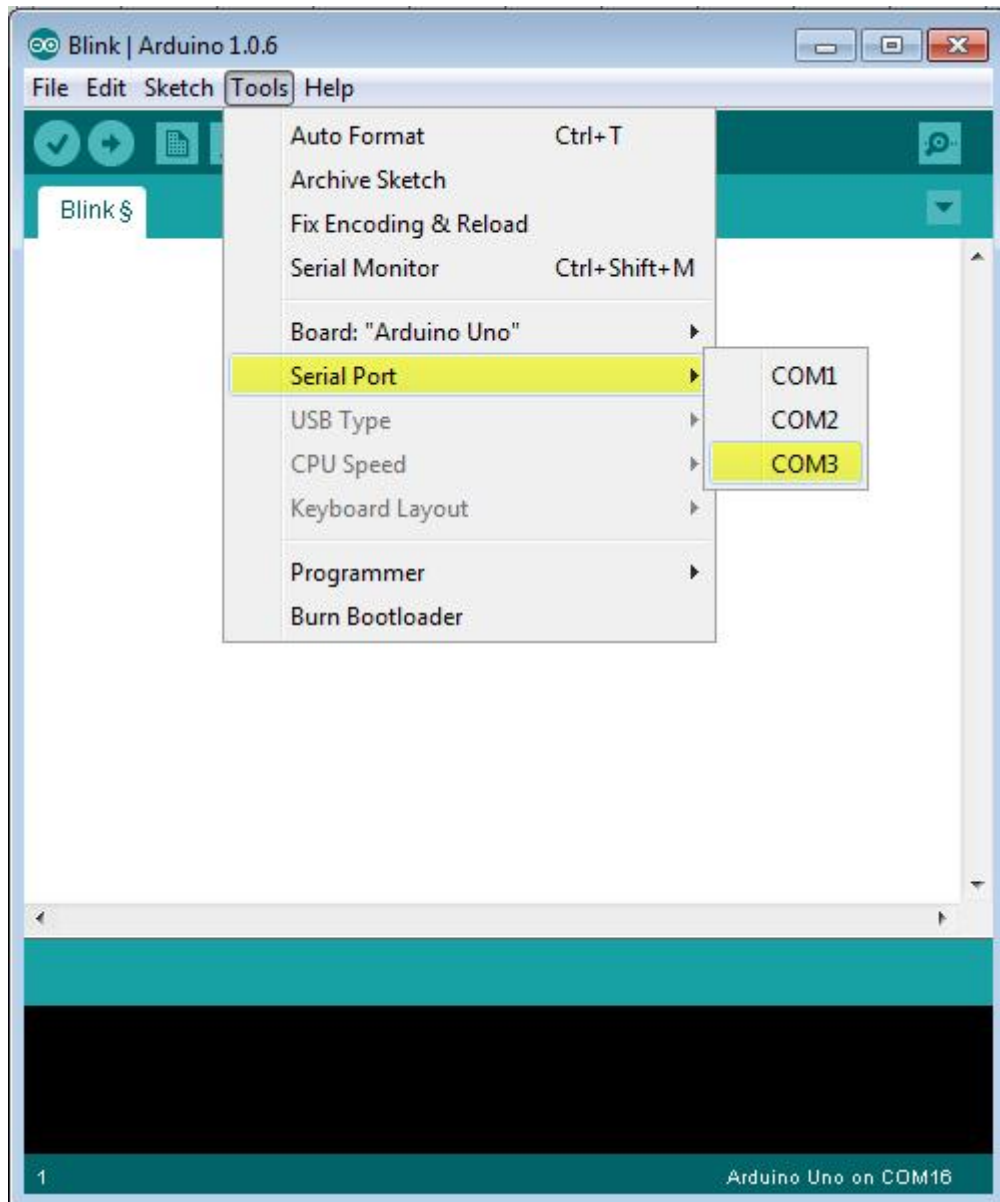


Fig 4.5: Selecting the Serial Port in the Arduino IDE

Step 8 – Upload the program to your board.

Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.

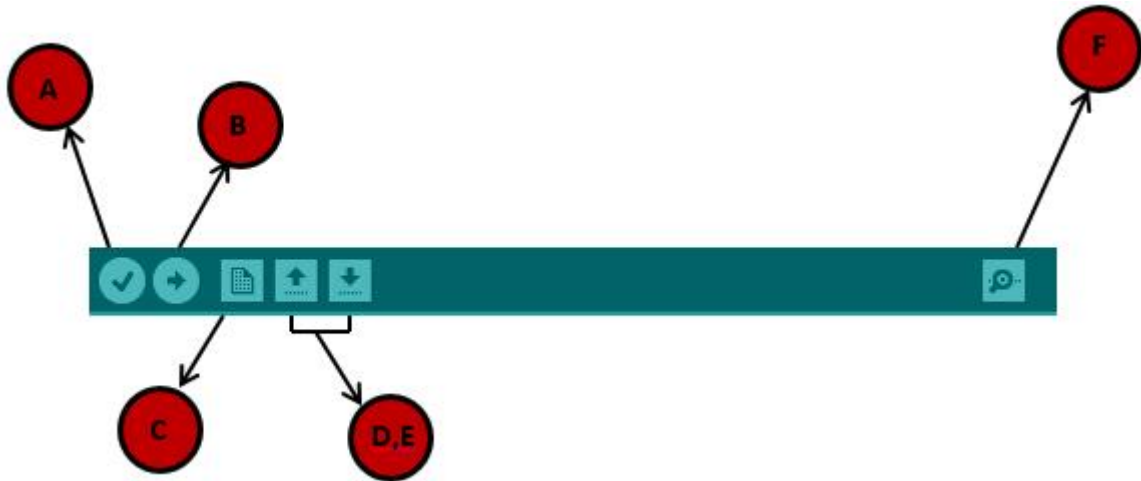


Fig 4.6:Arduino IDE Toolbar Icon

A – Used to check if there is any compilation error.

B – Used to upload a program to the Arduino board.

C – Shortcut used to create a new sketch.

D – Used to directly open one of the example sketch.

E – Used to save your sketch.

F – Serial monitor used to receive serial data from the board and send the serial data to the board.

Now, simply click the "Upload" button in the environment. Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.

Note – If you have an Arduino Mini, NG, or other board, you need to press the reset button physically on the board, immediately before clicking the upload button on the Arduino Software.

4.3 CONNECTING A BATTERY

For stand-alone operation, the board is powered by a battery rather than through the USB connection to the computer. While the external power can be anywhere in the range of 6 to 24 V (for example, you could use a car battery), a standard 9 V battery is convenient. While you could jam the leads of a battery snap into the Vin and Gnd connections on the board, it is better to solder the battery snap leads to a DC power plug and connect to the power jack on

the board. A suitable plug is part number 28760 from www.jameco.com. Here is what this looks like.

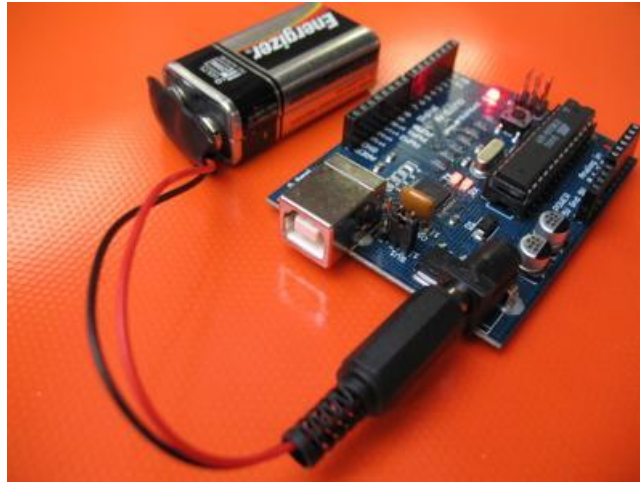


Fig 4.7:Arduino with battery

Disconnect your Arduino from the computer. Connect a 9 V battery to the Arduino power jack using the battery snap adapter. Confirm that the blinking program runs. This shows that you can power the Arduino from a battery and that the program you download runs without needing a connection to the host PC .

Moving On

Connect your Arduino to the computer with the USB cable. You do not need the battery for now. The green PWR LED will light. If there was already a program burned into the Arduino, it will run.

Start the Arduino development environment. In Arduino-speak, programs are called “sketches”, but here we will just call them programs.

In the editing window that comes up, enter the following program, paying attention to where semi-colons appear at the end of command lines.

```
void setup()
{
  Serial.begin(9600);
  Serial.println("Hello World");
}
void loop() {}
```



Fig 4.8:Arduino IDE with Hello World Program

4.4 MC PROGRAMMING LANGUAGE: EMBEDDED C -

This is the most widely used programming language for embedded processors/controllers. Assembly is also used but mainly to implement those portions of the code where very high timing accuracy, code size efficiency, etc. are prime requirements. Embedded C is perhaps the most popular languages among Embedded Programmers for programming Embedded Systems. There are many popular programming languages like Assembly, BASIC, C++ etc. that are often used for developing Embedded Systems but Embedded C remains popular due to its efficiency, less development time and portability.

CHAPTER-6

ADVANTAGES & APPLICATIONS

5.1 ADVANTAGES

- **Hands-Free Operation**
Appliances can be controlled using voice commands, eliminating the need for physical switches.
- **Multilingual Support**
Supports voice commands in English, Telugu, and Hindi, making it user-friendly for people of different language backgrounds.
- **Low Cost System**
Uses easily available components like Arduino and Bluetooth, making it economical compared to Wi-Fi or IoT systems.
- **Easy to Use**
Simple mobile application and voice control make the system easy for all users.
- **Wireless Control**
Bluetooth provides reliable short-range wireless communication without complex setup.
- **Helpful for Elderly and Physically Challenged People**
Voice control reduces physical effort and improves accessibility.
- **Real-Time Feedback**
LCD display shows the current status of appliances, ensuring clear confirmation.
- **Low Power Consumption**
Bluetooth consumes less power compared to Wi-Fi-based systems.
- **Safe Operation**
Relay modules provide electrical isolation, ensuring user safety while controlling high-voltage appliances.
- **Expandable System**
More appliances can be added easily by modifying hardware and software.

5.2 APPLICATIONS

- Smart Home Automation

Used to control home appliances like lights and fans using voice commands.

- Energy Management

Appliances can be turned OFF easily, helping to save electricity.

- Small Offices & Rooms

Can be used in offices, hostels, or classrooms for simple automation

- Educational Projects

Widely used by students for learning Arduino, Bluetooth, and automation.

5.3 LIMITATIONS

- Limited Range

Bluetooth works only within ~10 meters.

- No Remote Access

Cannot control devices from far locations.

- Depends on Smartphone

System works only when mobile phone is available.

- Voice Recognition Issues

Errors may occur due to noise or unclear speech.

- Limited Device Control

Only a few appliances can be connected unless hardware is expanded.

- Security Weakness

Bluetooth is less secure compared to Wi-Fi/IoT systems.

- Power Failure Problem

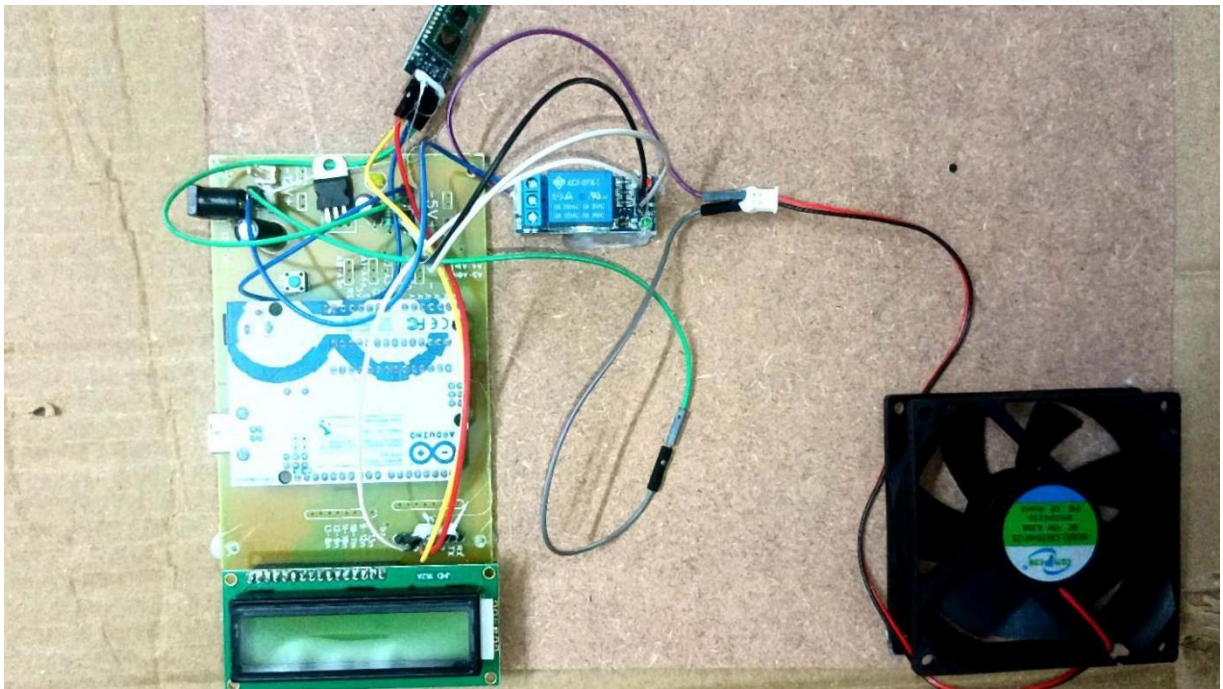
System stops working if power supply is lost.

- Language Limitations

Only supports selected languages (not universal).

CHAPTER-6

RESULT



CHAPTER-7

CONCLUSION

This project demonstrates a Bluetooth-based home automation system using multilingual voice commands. Users can control appliances like lights and fans through a smartphone. The system integrates Arduino, Bluetooth, relays, and an LCD for efficient operation. It supports multiple languages (English, Telugu, and Hindi), making it accessible and user-friendly. The system is low-cost, easy to implement, and works reliably within Bluetooth range.

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